Anthony Lavertu

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HIGHLIGHTS

- Closed-loop ML/wet-lab AMP discovery system (ML lead): **25% anticipated hit rate**.
- **3x first-author scientific contributions** (Scientific Reports, ASMS poster, technical report targeting ICLR).
- **NSERC + FRQNT** funded (twice).
- **4.33/4.33** (M.Sc.), **4.29/4.33** (B.Sc.).

RESEARCH EXPERIENCE

Graduate Researcher — Corbeil Lab *Université Laval, Summer 2025 – Present*

- **Lab-in-the-loop for AMP discovery (ML lead):** built and led the ML side of a closed-loop AMP discovery project with a wet-lab collaborator running a 10K-sample high-throughput potency assay. Trained a fine-tuned transformer with a custom regression head to predict log-fold-change activity. Leveraged it to annotate a database of 870K peptides, and used the model's top and most uncertain predictions to select the next batch for testing (~25% hit rate expected on the first round, validation ongoing). Now building a pretraining scheme with stronger inductive bias for bioactive peptides, and exploring generative models (diffusion, GFlowNets) to design candidates directly instead of just screening them.

- **QMAP Benchmark:** discovered that random train/test splitting causes severe data leakage in AMP potency prediction, and that standard homology-clustering methods (CD-HIT, MMseqs2) only partially fix it; built a similarity-graph and community-detection-based splitting method to prevent leakage, and a benchmark enforcing it. Results on the benchmark show limited real progress and poor generalization in the field over six years, despite reported gains under weaker evaluation (first-author paper, Scientific Reports 2026).

- **Refnd:** generalized the QMAP splitting method beyond AMPs into a formal framework for non-i.i.d. biochemical data, and derived an O(n log n) approximation algorithm from it — cutting split times from hours to seconds versus existing methods, and turning splits that would take months on large datasets into hours. It is significantly more robust than other approaches, having never failed yet, whereas other methods regularly fail on some hard datasets (validated on 10+ datasets across 4 modalities). Under review on arXiv, targeting ICLR 2027.

- **MALDI deconvolution:** co-developed a probabilistic model that recovers bacterial composition from polymicrobial MALDI-TOF spectra by maximum-likelihood decomposition into per-species reference signatures (first-author poster, ASMS 2026).

Research Collaborator — NORLAB *Université Laval, Fall 2025 – Present (part-time)*

- Developed a visual SLAM (Simultaneous Localization and Mapping) method robust to high-dynamic-range scenes, where standard auto-exposure saturates, losing the image features the system relies on and corrupting its position/map estimates. After a deep-learning optimal-exposure predictor proved unreliable, switched to capturing multiple fixed-exposure images per frame (e.g., AE20/AE50/AE80) and added custom factors (constraints) so exposure changes no longer destabilize the estimate. Started as a course project during the M.Sc. and grew into an ongoing collaboration.

Research Intern — Corbeil Lab *Université Laval, Summer 2024*

- Research internship funded by NSERC/FRQNT; built a deep learning AMP potency predictor, later found to be affected by the data-leakage issue that motivated QMAP.

Research Intern — FLC Lab (Flavie Lavoie-Cardinal) *Université Laval, Summer 2023*

- Applied active learning and anomaly detection to identify abnormal synapses in Super Resolution Microscopy images as potential early precursors of Parkinson's disease.

PUBLICATIONS & PRESENTATIONS

- Lavertu, A., Corbeil, J., Germain, P. **QMAP: A Benchmark for Standardized Evaluation of Antimicrobial Peptide MIC and Hemolytic Activity Regression.** Scientific Reports (Nature), 2026. Code: github.com/anthol42/QMAP
- Lavertu, A., Côté, J., Corbeil, J., Gobeil, S., Germain, P. **Refnd: Preventing Data Leakage in Relational Datasets.** Technical report, under review (arXiv); target venue: ICLR. github.com/anthol42/refnd
- **Probabilistic Deconvolution of Polymicrobial MALDI Mass Spectra Using Maximum Likelihood Decomposition.** Poster, ASMS Conference on Mass Spectrometry and Allied Topics, San Diego, CA, May 31 – Jun 4, 2026 (Abstract ID 326812).

AWARDS & FUNDING

- **NSERC** — Canada Graduate Scholarship – Master's Program (M.Sc.)
- **FRQNT** — Bourses de maîtrise en recherche (M.Sc.)
- **NSERC** — Undergraduate Student Research Award (2nd research internship)
- **FRQNT** — Supplément aux bourses de 1er cycle du CRSNG – Académique, BPCA (2nd research internship)

EDUCATION

M.Sc., Computer Science *Université Laval, Summer 2025 – Present*

Maîtrise en informatique – avec mémoire.

Cumulative average: 4.33/4.33. NSERC- and FRQNT-funded.

B.Sc., Bioinformatics *Université Laval, 2022 – 2025*

Baccalauréat en bio-informatique.

Cumulative average: 4.29/4.33.

LEADERSHIP & EXTRACURRICULAR

Team Lead, Autonomous FPV Drone Laser-Tag Competition *AI Club, Université Laval, 2024 – Present*

Led a student team building autonomous FPV drones that compete in a laser tag game. Worked on hardware, perception, control, and autonomy stack.

Member, iGEM Team *Université Laval, 2023 – 2024*

Contributed literature review and bioinformatics work to a synthetic biology research project for the international competition.

SKILLS

ML/DL: Pretraining, LLMs, PLMs, GFlowNets, Active Learning, Anomaly Detection, Fine-Tuning, Graphs
Bioinformatics: UniProt, PDB, ESM Atlas, DBAASP, AlphaFold, MMseqs2, etc.; sequence/structure data pipelines, wet-lab collaboration workflows
Other: Scientific Writing, Literature Review, Independent Project Ideation and Execution
Languages: Bilingual, French and English

OPEN-SOURCE TOOLS

- **deepboard** (github.com/anthol42/deepboard) — a more flexible, modern alternative to TensorBoard for logging and browsing deep learning experiment results locally.
- **torchbuilder** (github.com/anthol42/torchbuilder) — a CLI tool that scaffolds PyTorch project structure and boilerplate.